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----- Active space -----

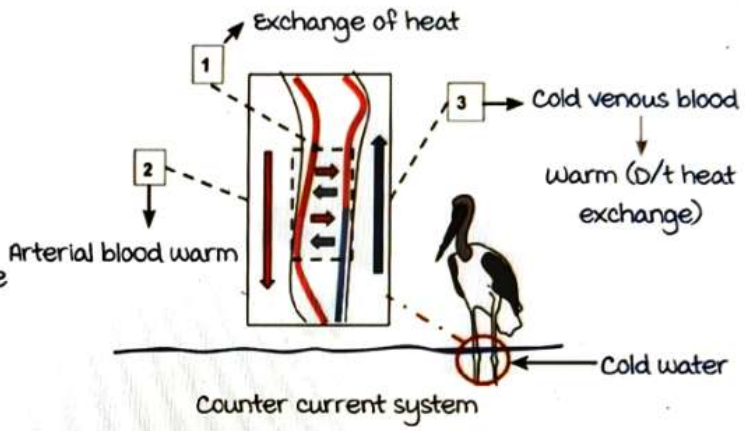
LOOP OF HENLE AND DISTAL CONVOLUTEI TUBULE

Counter Current Mechanism

00:00:25

Requirements for counter current system :

1. 2 tubes lying close to each other.
2. Fluid movement in opposite directions in the tubes.



Seen in :

- Human Kidneys (Loop of henle).
- Testis.
- Intestines.
- Skin (Temperature regulation).

Loop of henle (LOH)

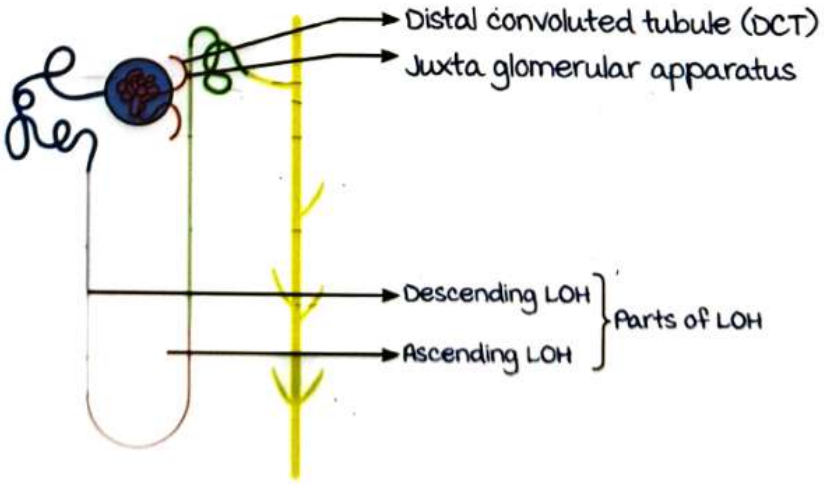
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Loop of henle receives isotonic fluid from PCT (Osmolarity = 300 mosm/l).

Function : Concentration and dilution of urine.

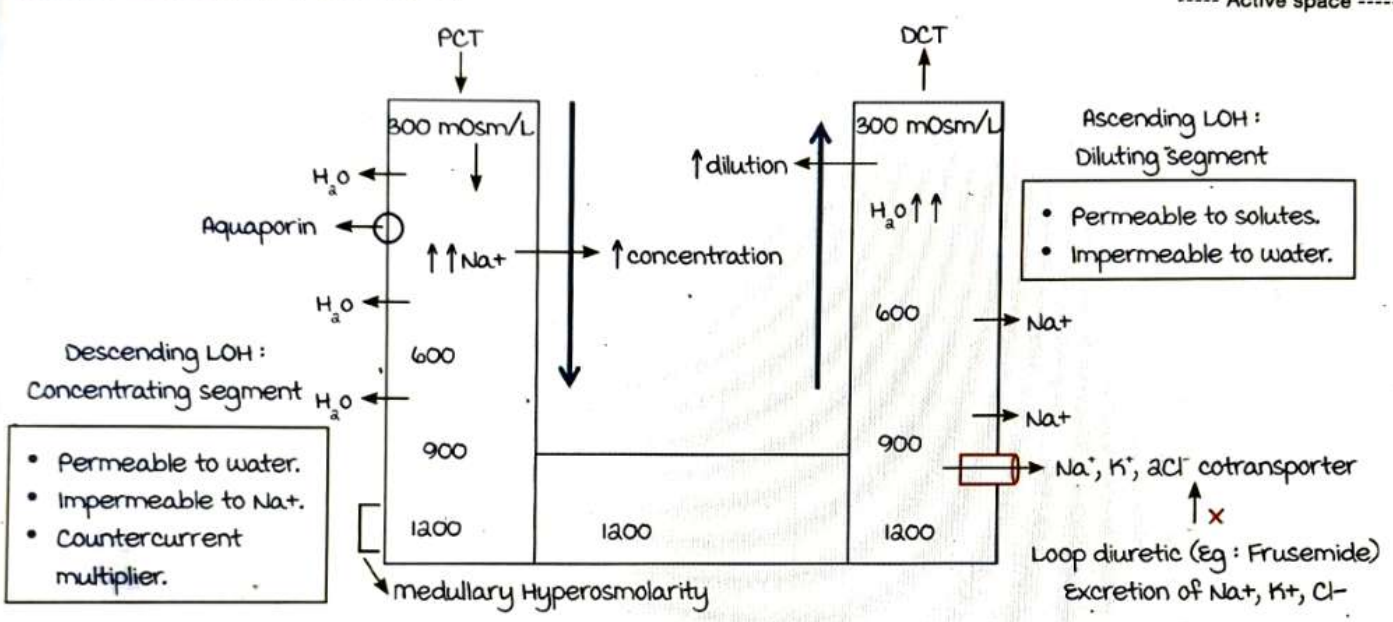
Range of urine osmolarity = 50-1400 mosm/L.

Average = 100-800m Osm/L.



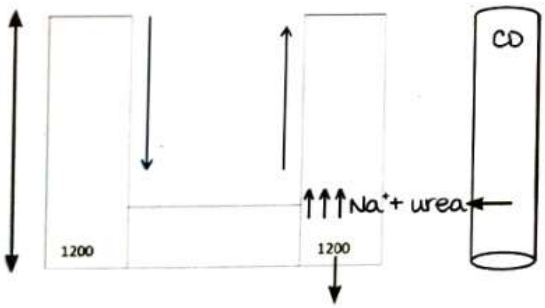
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COUNTER CURRENT SYSTEM :



Note: Single effect → Transport of Na+ without being accompanied bu H O

medullary hyperosmolarity:



- Created by : Countercurrent multiplier.
- Purpose : ADH mediated water reabsorption from CD.
- Maintained by : Countercurrent exchanger (Vasa recta).

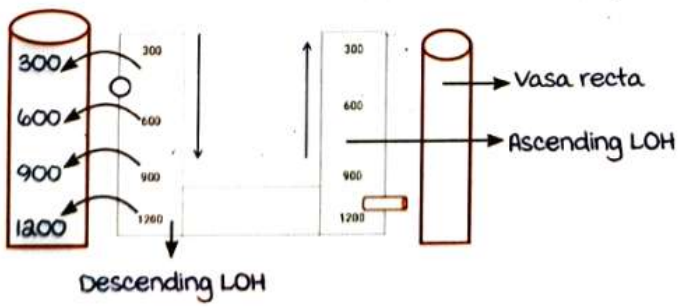
Length of LOH α medullary hyperosmolarity

Contributed by :

- Na+ (50%).
- Urea (50%) : From CD.
  - By urea transporters (UT1 and UT2.)
  - ↑ with high protein diet.

vasa recta :

Blood vessels along LOH of similar osmolarities

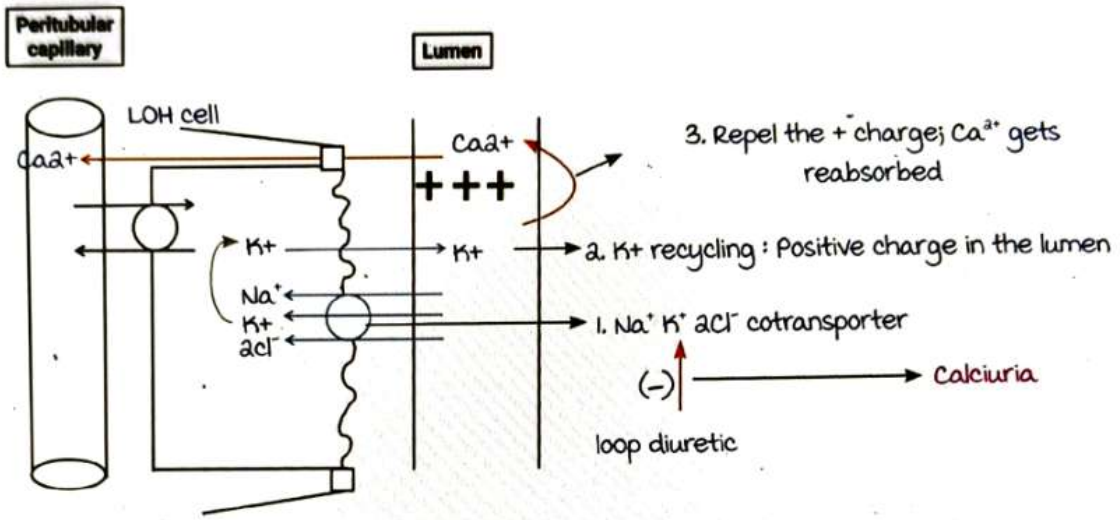


Feedback

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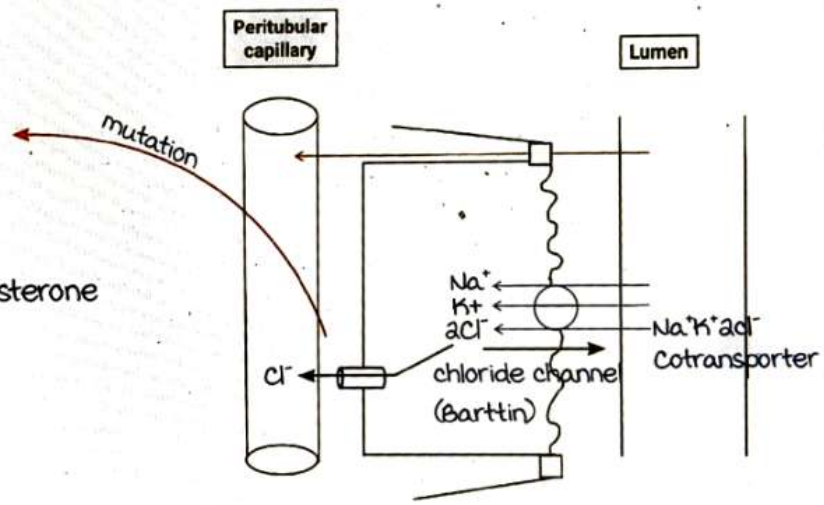
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REABSORPTION OF Ca2+



REABSORPTION OF CHLORIDE (CL^-)

Bartter syndrome :
Salt & water loss
(Volume depletion)
↓
RAAS activation : ↑ Aldosterone
↓
↑ K+ & H+ excretion

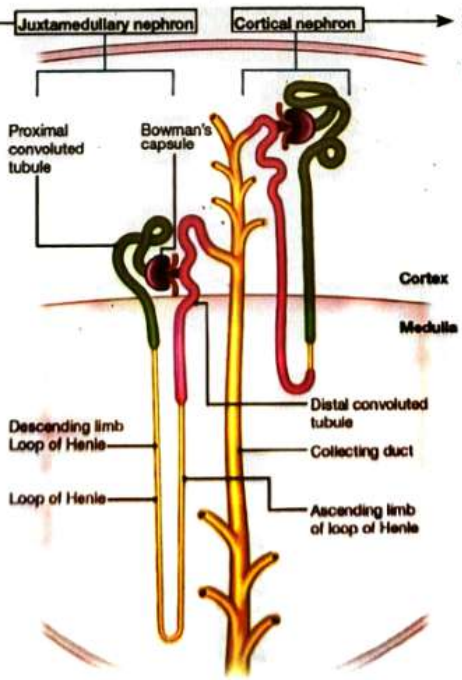


CLASSIFICATION BASED ON LENGTH

Long LOH ← Juxtamedullary nephron | Cortical nephron → Short LOH :

- 15% of total nephrons.
- Responsible for conc. & dilution of urine.

85% of total nephrons (maximum)



Feedback

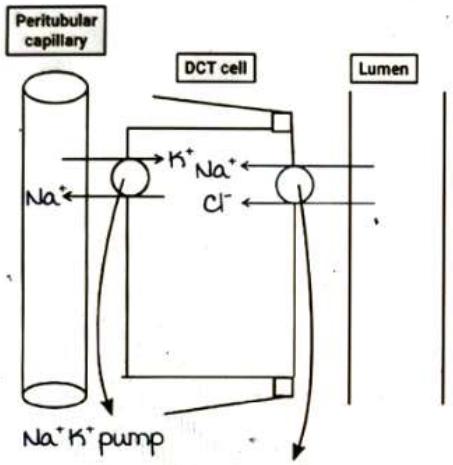
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Distal Convoluted Tubule

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Sodium reabsorption :

5% of filtered load.

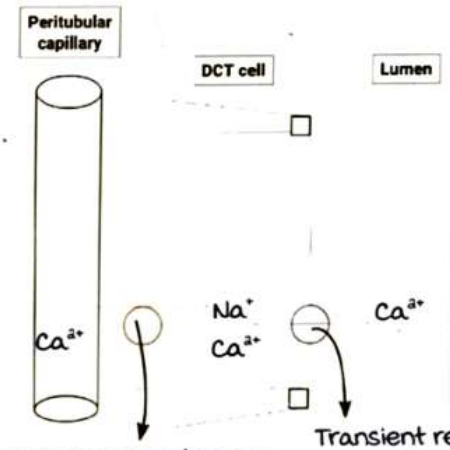


Na+-Cl- cotransporter (2° active transport) → Mutation → Gitelman's syndrome

(-) Thiazide diuretics

Calcium reabsorption :

9% of filtered load.



Transient receptor potential vanilliod channel (TRPV 5):
up regulated by parathormone (PTH) and vitamin D

Juxta Glomerular Apparatus

00:34:35

DCT coming in close contact to its glomerulus.
Forms juxtaglomerular apparatus.

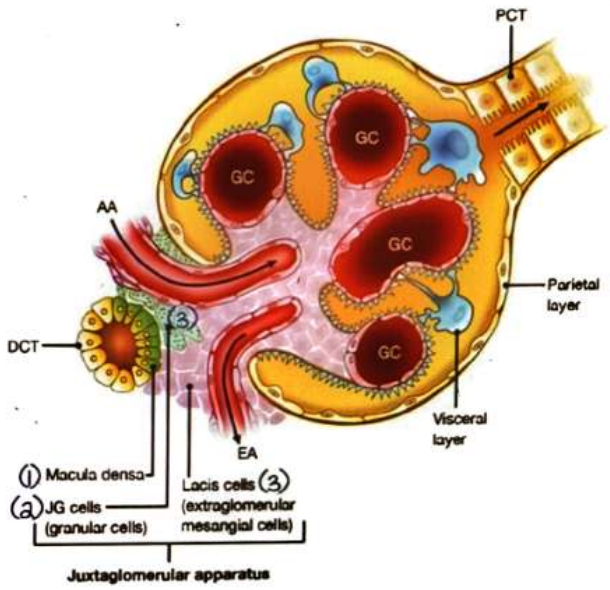
TYPES OF CELLS :

1. JG cells:

- Line the afferent arteriole.
- AKA granular cells; contain granules → Release renin.
- JG cells sense pressure changes.

Conditions that ↑ renin release :

- Low pressure/hypotension in afferent arteriole.
- ↓ Na+.
- Sympathetic stimulation.
- Diuretics.
- Hemorrhage.



Feedback

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----- Active space -----

Conditions that ↓ renin release :

- ↑ Pressure in afferent arteriole.
- Angiotensin II (–ve feedback).
- ADH.

Renin-angiotensin-aldosterone system (RAAS) : In response to low BP

Angiotensinogen (Liver) —Renin→ Angiotensin I —ACE in lung→ Angiotensin II → Aldosterone from adrenal cortex

Angiotensin II → ↑ BP

Aldosterone from adrenal cortex → ↑ Na+ reabsorption

Overactive RAAS : Hypertension (HTN).

- In treatment of HTN, ACE inhibitors are used to block RAAS.
- Direct renin blockers : Aliskiren.

2. macula densa :

- GFR/NaCl sensors : Chemical sensor.
- Function → Tubuloglomerular feedback (TGF) :

↑ GFR → ↑ NaCl filtration → Sensed by macula densa in distal tubule → Releases adenosine → Constriction of afferent arteriole ↓ GFR

3. Extraglomerular mesangial cells/lacis cells/Polkissen cell :

Supporting cells.

Feedback